

ISEN/DAEN 427

Decision Making and Risk



and finish a marathon (*why not*)
become a fantastic professor
prepare/publish my textbook
see my son grow up



Instructional Assistant Professor, faculty senator. 2024

postdoc TAMU/NIST Risk Community Planning 2002-2023

PhD in **Urban and Regional Science**. 2022

start teaching research methods, statistics, and GIS. 2017

start working with hurricane evacuation studies, 2015

move to **College Station**, 2007

master in **Production Engineering**. 2006

work with **forecast models and logistics**. 2002-2005

graduate in **Civil Engineering**. 2001

born in Rio, 1976



Street Games (Echmiadzin, Armenia)
Maciej Dakowicz 2017.07.06

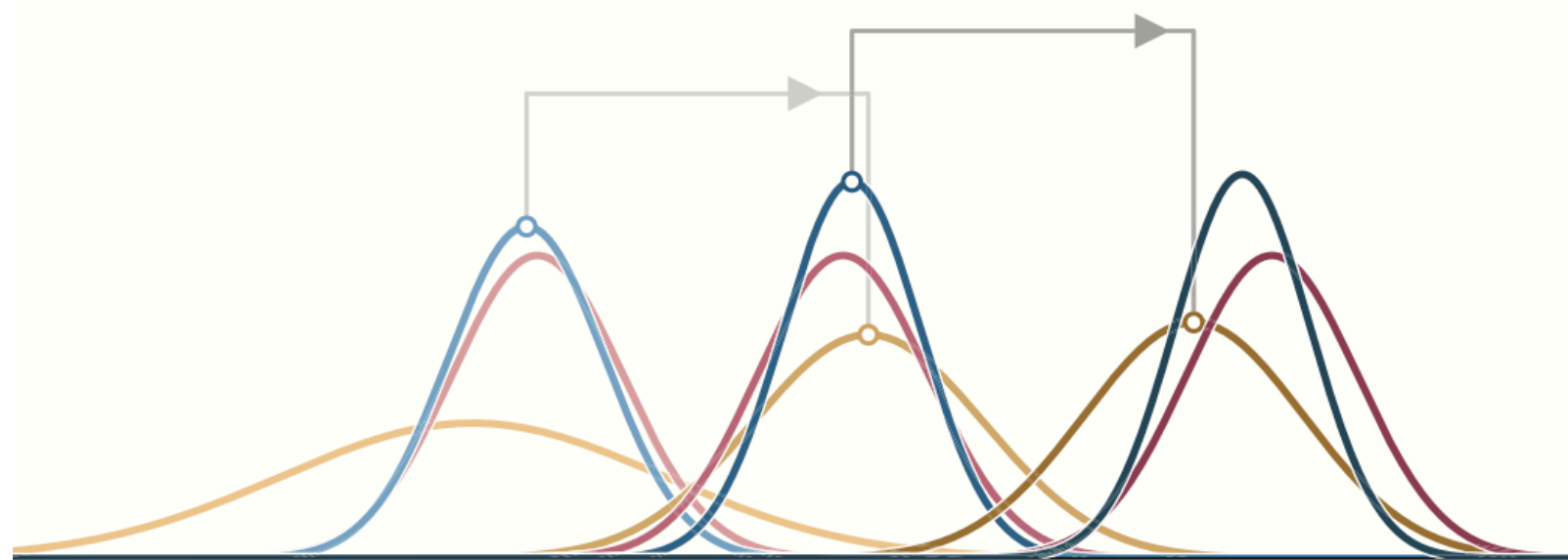


- Review the syllabus & calendar
- **Keep calm, enjoy the semester!**
- **And come to office hours!**

Textbooks

Bayesian Models of Perception and Action

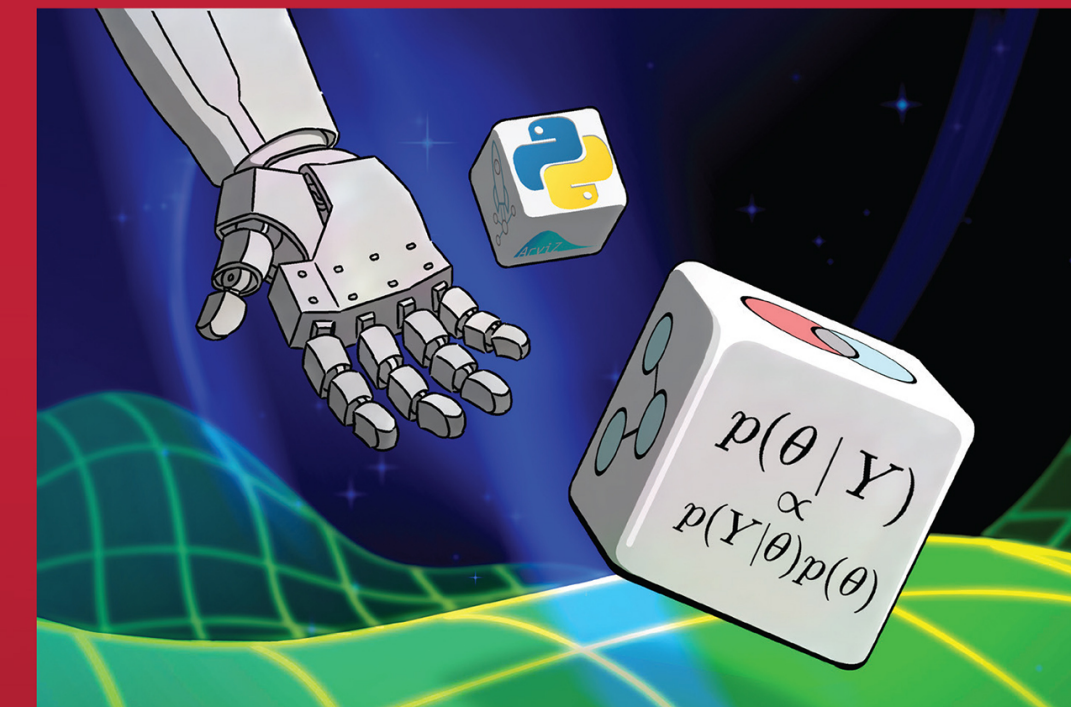
An Introduction



Wei Ji Ma
Konrad Paul Kording
Daniel Goldreich

Texts in Statistical Science

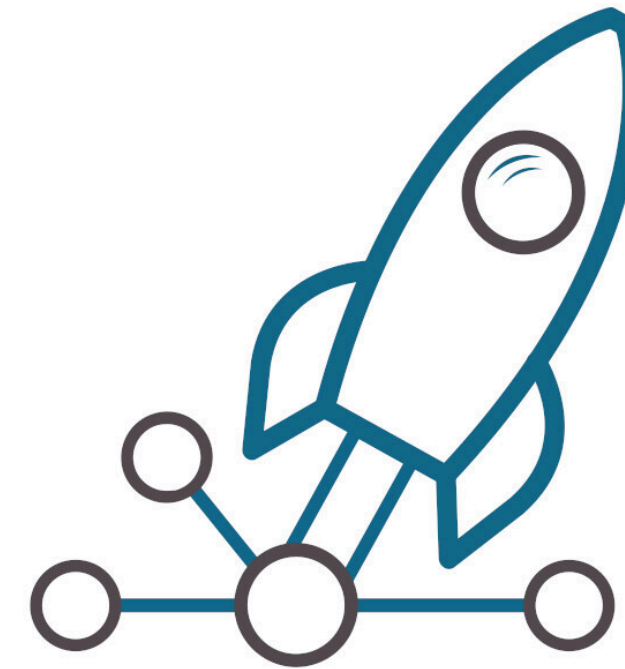
Bayesian Modeling and Computation in Python



Osvaldo A. Martin
Ravin Kumar
Junpeng Lao

 CRC Press
Taylor & Francis Group
A CHAPMAN & HALL BOOK

Class technology



PYMC



Sucking

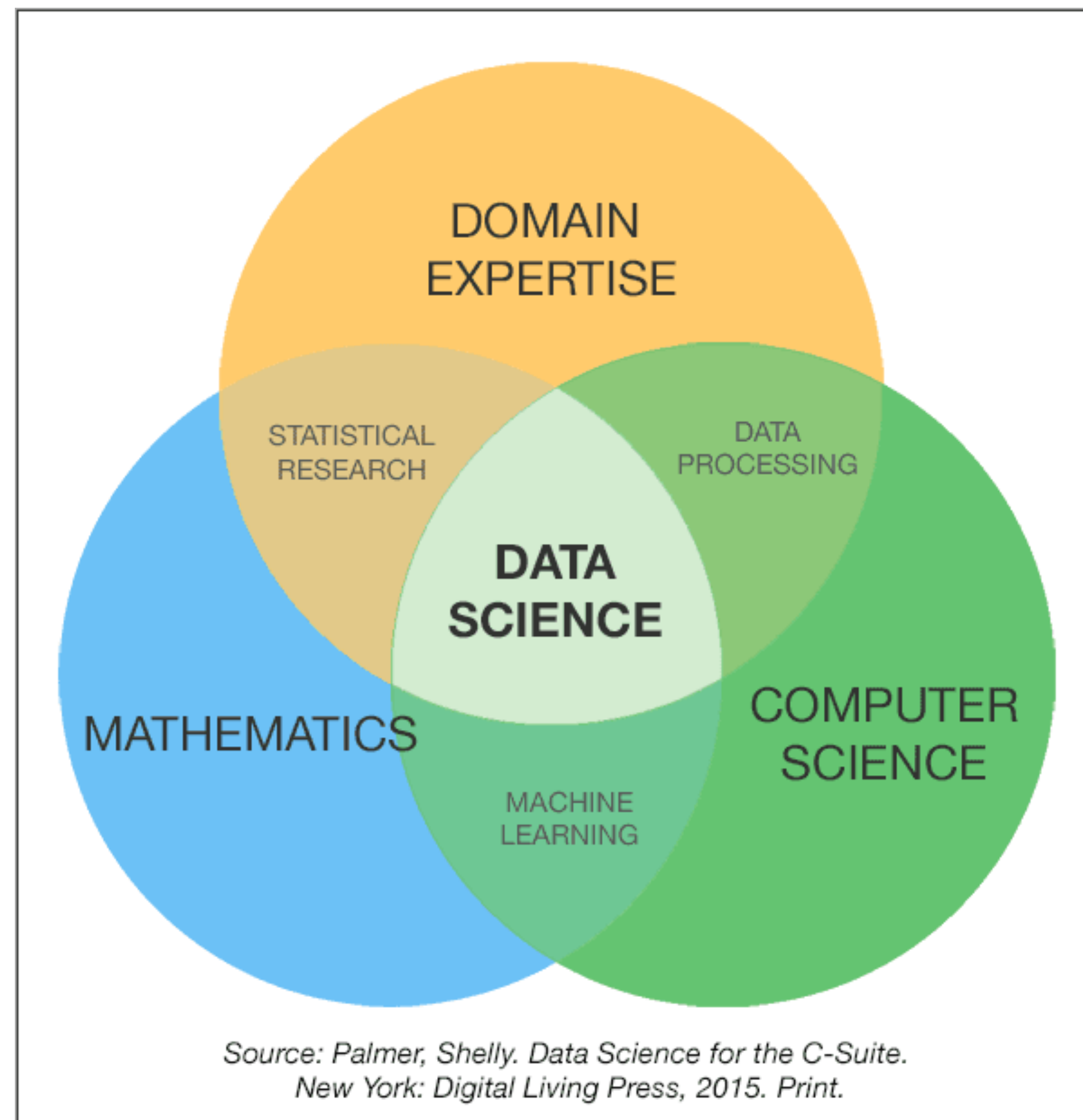


"There is no way of knowing nothing about a subject to knowing something about a subject without going through a period of much frustration and suckiness."

"Push through. You'll suck less."

Hadley Wickham, author of ggplot2

Am I making you data engineer?

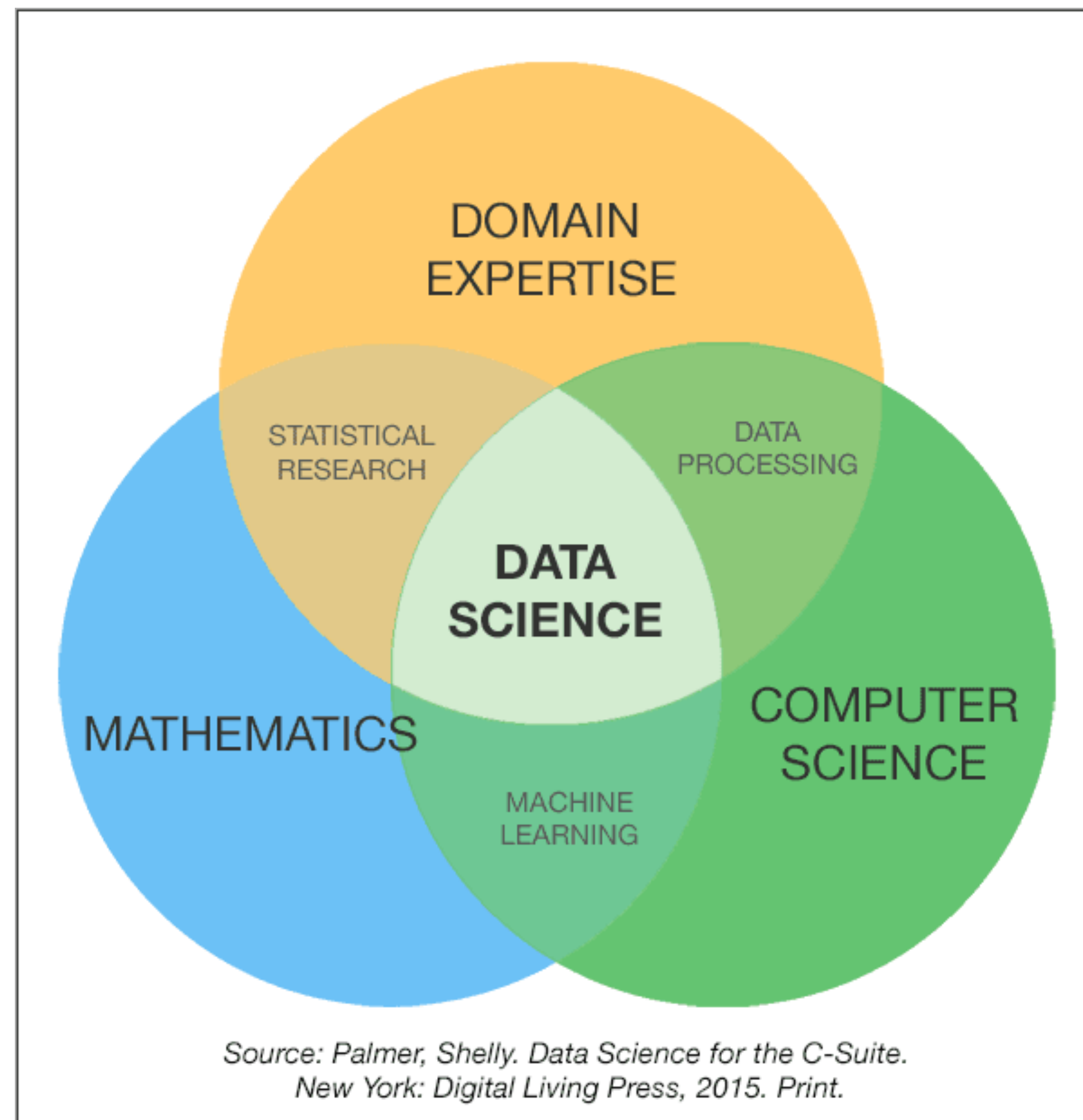


Data science
(data + statistics + communication)

Causal inference
(DAGs + stats)

You don't need to be a mechanic to drive a car safely

Am I making you data engineer?



A simple analogy, cooking:

- **data** is like **individual ingredients**, flour, sugar, eggs.
- **information** is like a **baked cake**, ingredients combined and structured into something meaningful.

Goals for the class

Draw on concepts from economics, psychology, and management science to analyze decision-making processes

Understand Bayesian approach to perception and decisions under uncertainty

Implement computational models for Bayesian inference

~~Become an expert ...~~

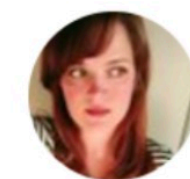
It would take 10,000 hours

Goals for the class

Draw on concepts from economics, psychology, and management science to understand decision making processes

Understand perception

Implement concepts



Jesse Maegan
@kierisi

Following

My #rstats learning path:

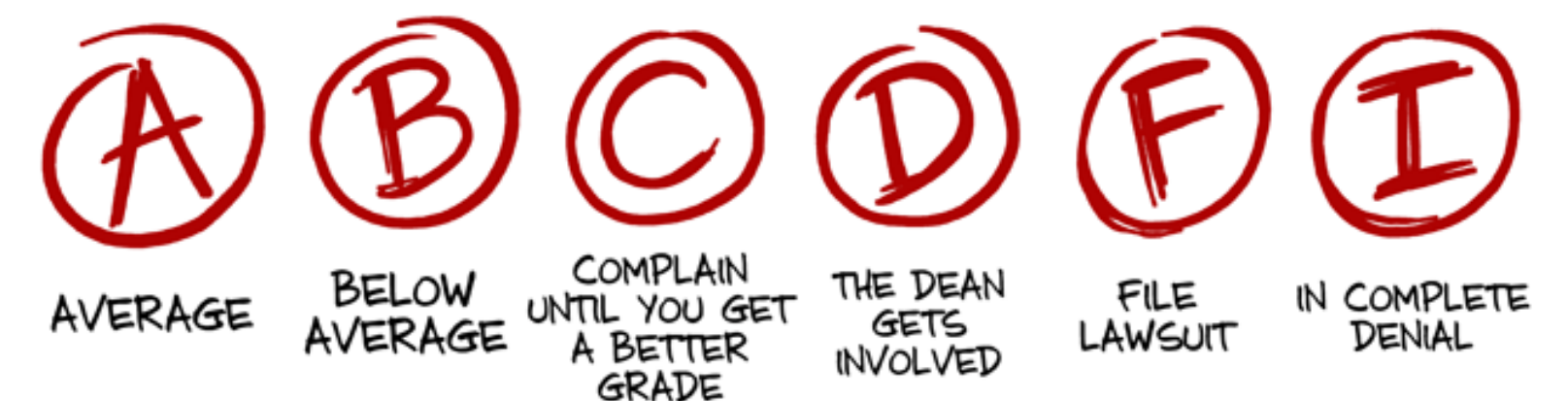
1. Install R
2. Install RStudio
3. Google "How do I [THING I WANT TO DO] in R?"

Repeat step 3 ad infinitum.

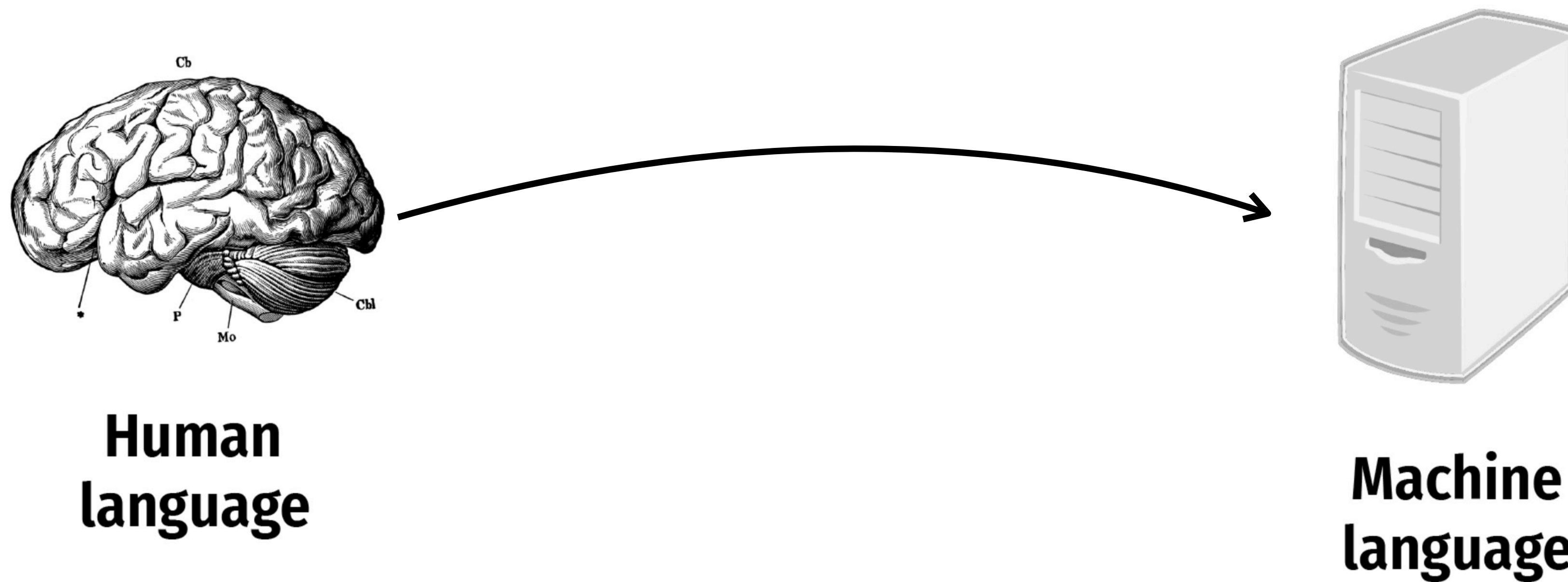
7:19 AM - 18 Aug 2017

~~Become an expert ...~~

GRADE INFLATION



Goals for the class

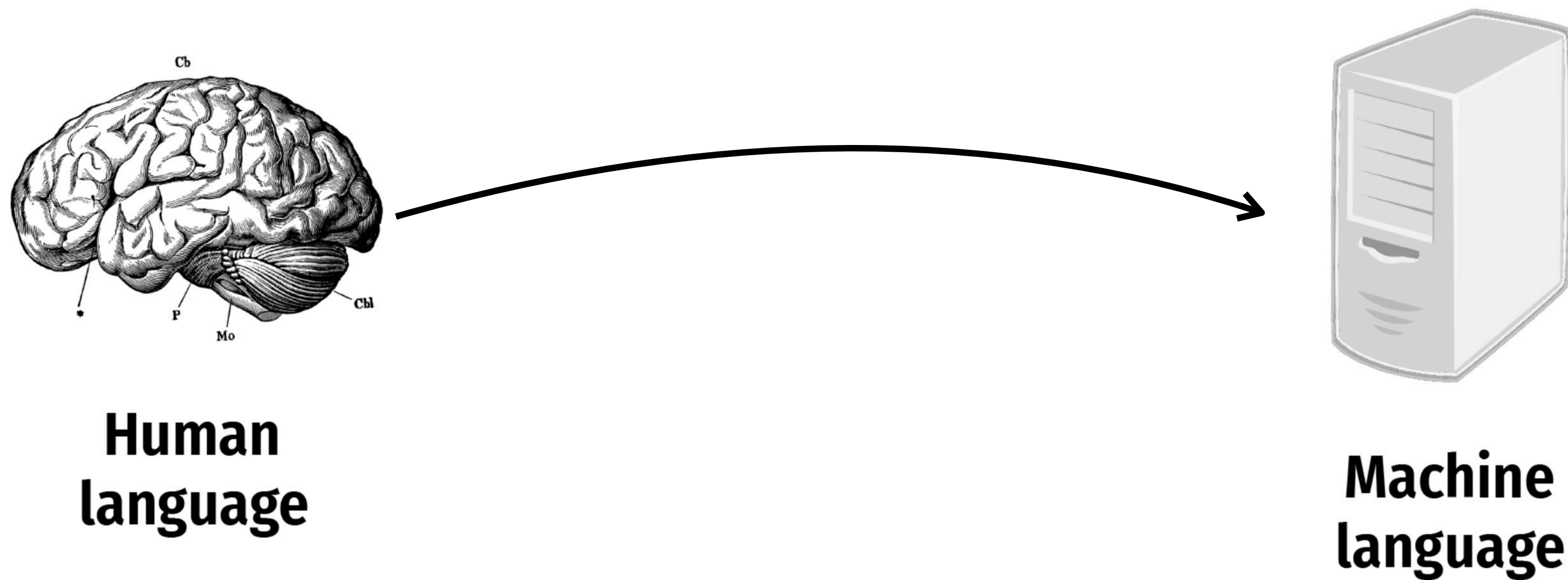


Speak and do causation

Design rigorous evaluations

Change the world
with data

Goals for the class



In lists of key employability skills, **problem solving** usually appears at the top (makes sense). What is the difference between problem-solving and problem-structuring?

The key to answering this question is not in a protracted wrangle about the differences but in examining the nature of problems problem-structuring involves **defining, organizing, and clarifying** a problem before attempting to solve it, ensuring a more effective resolution!